

A Finite Local Koopman Jet for a Polynomial Hénon Germ

Anonymous Route-A report

Abstract

We form the finite pullback of $F(u, v) = (u^2 + 3u/2 - v/2, u)$ on $\mathbb{Q}[u, v]/(u, v)^5$. Exact substitution gives a fifteen-dimensional rational matrix with trace $129/16$ and determinant 2^{-20} . This baseline report records a finite local calculation only.

1 Frozen quotient

Let $\mathfrak{m} = (u, v)$ and $A_4 = \mathbb{Q}[u, v]/\mathfrak{m}^5$. We use the degree-ordered basis

$$(1; u, v; u^2, uv, v^2; u^3, u^2v, uv^2, v^3; u^4, u^3v, u^2v^2, uv^3, v^4).$$

For $K[p] = [p \circ F]$, truncated rational expansion constructs all entries of a 15×15 matrix. Its computed characteristic data factor as

$$\chi_K(\lambda) = \prod_{k=0}^4 (\lambda - 2^{-k})^{5-k}, \quad \det(I - zK) = \prod_{k=0}^4 (1 - 2^{-k}z)^{5-k}.$$

Thus $\text{tr } K = 129/16$ and $\det K = 2^{-20}$.

2 Baseline boundary

The operator is only the finite local quotient just printed. This draft does not provide a proof from the maximal-ideal filtration, a matrix convention, or a nonlinear-versus-linear control audit. It makes no claim of a global Koopman spectrum, nuclearity, a Fredholm determinant, arithmetic data, a Hilbert–Pólya operator, or Route B. The release scope is NO_BAD_EULER_OR_ROOT_NUMBER.